Project Conclusion: Video Game Market Segmentation

1. Project Overview

This Data Mining project applied **unsupervised machine learning techniques** to segment the global video game market into meaningful, actionable clusters. The goal was to discover hidden patterns and natural groupings within video game sales data that could inform strategic business decisions for developers, publishers, and investors.

2. Methodology Summary

The project followed a comprehensive data mining pipeline:

Phase	Description	Key Techniques
1. Data Preprocessing	Cleaning, imputation, feature engineering	Median imputation, RobustScaler normalization, 51 engineered features
2. Clustering Analysis	Algorithm comparison and optimization	K-Means (K=2-6, 200 restarts), DBSCAN (eps: 3.77-4.89)
3. Evaluation	Cluster quality assessment	Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Score
4. Business Interpretation	Segment characterization and recommendations	Success factor analysis, competitive landscape assessment |

3. Feature Engineering Details (51 Features)

The 51 engineered features were designed to capture different aspects of video game market positioning.

3.1. CORE FEATURES (17 features)

total_sales, na_sales, *jp_sales, pal_sales, other_sales, critic_score, game_age, na_sales_ratio, jp_sales_ratio, pal_sales_ratio, other_sales_ratio*`

sales_concentration_hhi: Herfindahl-Hirschman Index for regional concentration

global_game: Binary flag: 1 if sales distributed globally

regional_specialist: Binary flag: 1 if >70% sales from one region

is_hit: Binary flag: 1 if total_sales > threshold (hit game)

pub_avg_sales: Publisher's historical average sales per game

pub_game_count: Total games published by this publisher

3.2. GENRE FEATURES (14 features): binary type

genre_sports, genre_action, genre_misc, genre_shooter, genre_role_playing, genre_racing, *genre_adventure*, *genre_platform*, *genre_simulation*, *genre_fighting*, *genre_popularity*, *genre_avg_quality*, *genre_avg_sales*, *genre_competition*

3.3. **PLATFORM FEATURES (10 features):** binary type

platform_nintendo, *platform_sony*, *platform_microsoft*, *platform_pc*, *platform_sega*, *console_generation*, *platform_popularity*, *platform_avg_sales*, *platform_avg_quality*,*is_handheld*

3.4. INTERACTION FEATURES (10 features):

quality_per_sales: Critic score divided by sales (quality efficiency)

sales_per_quality: Sales divided by critic score (commercial efficiency)

sales_vs_quality_ratio: Ratio comparing sales performance to quality

dominant_region_strength: How dominant the strongest region is

dominant_region_id: Which region is dominant (NA=0, JP=1, PAL=2, Other=3)

regional_balance: 1 - HHI (degree of global diversification)

pub_genre_expertise: Publisher's expertise in this specific genre

year_competition: Number of competing games released same year

age_sales_product: Game age × sales interaction

platform_genre_fit: Historical success of this genre on this platform

3.5. Most Important Features for Clustering Decision

The K-Means algorithm uses **all 51 features** for clustering, but some features have **higher impact** on separating the clusters:

Rank	Feature	Why It Matters for Clustering
1	*total_sales*	**Primary differentiator** - directly separates Premium (high sales) from Mass Market (low sales)
2	*sales_concentration_hhi*	Distinguishes **global games** (low HHI, Premium) from **regional specialists** (high HHI, Mass Market) |
3	*is_hit*	*is_hit*
4	*pub_avg_sales*	Premium games tend to come from **publishers with higher historical success**
5	*regional_balance*	Premium games have **more balanced global distribution** across NA, JP, EU
6	*genre_sports*	Sports games are **overrepresented in Premium segment** (higher average sales)
7	*platform_sony*	PlayStation platforms **correlate with Premium segment** performance
8	*critic_score*	Higher quality scores **weakly correlate** with Premium segment membership
9	*na_sales_ratio*	Premium games have **higher NA market penetration** on average |
10	*pub_game_count*	Larger publishers (more games) tend to have **more Premium titles**

3.6. How Clustering Uses These Features:

Cluster 0 (Premium Performers):
→ High total_sales (avg $1.30M)
→ Low sales_concentration_hhi (globally distributed)
→ is_hit = 1 (98% are hits)
→ High pub_avg_sales (successful publishers)
→ High regional_balance (balanced NA/JP/EU sales)

Cluster 1 (Mass Market):
→ Low total_sales (avg $0.24M)
→ Higher sales_concentration_hhi (regional focus)
→ is_hit = 0 (mostly non-hits)
→ Lower pub_avg_sales (varied publishers)
→ Lower regional_balance (often regional specialists)

3.7. Feature Scaling: All 51 features were normalized using **RobustScaler** before clustering to ensure no single feature dominates due to scale differences.

4. PCA Analysis and Impact on Silhouette Score

4.1. **Principal Component Analysis (PCA)** was applied for dimensionality reduction and cluster validation:

Configuration	Value
Components	2 (for visualization)
Variance Explained	~45-50% of total variance
Purpose	Visualization and alternative evaluation

4.2. **Silhouette Score Comparison: Original Features vs PCA:**

K Value	Original Features	PCA (2 Components)	Winner
K=2	**0.4698**	0.4432	Original ✓
K=3	0.3845	0.3621	Original ✓
K=4	03512	03298	Original ✓
K=5	0.3201	0.3045	Original ✓

4.3. **Key Finding:** Original features consistently outperformed PCA-reduced features across all K values. This indicates that:

- **Information Loss:** PCA's 2 components captured only ~50% variance, losing important clustering signals
- **Feature Richness:** The 51 engineered features contain meaningful distinctions that benefit from full dimensionality
- **Decision:** Final clustering used original scaled features (not PCA) for optimal results
- **PCA Role:** Used primarily for 2D visualization, not for final cluster assignment

5. Key Results

5.1. **Clustering Performance:**

- **Algorithm Selected:** K-Means Clustering
- **Optimal Clusters:** 2 distinct market segments
- **Silhouette Score:** 0.4698 (approaching target threshold of 0.50)
- **Total Games Analyzed:** 12,937 commercial video games
- **Features Used:** 51 engineered features (sales, quality, regional, publisher metrics)

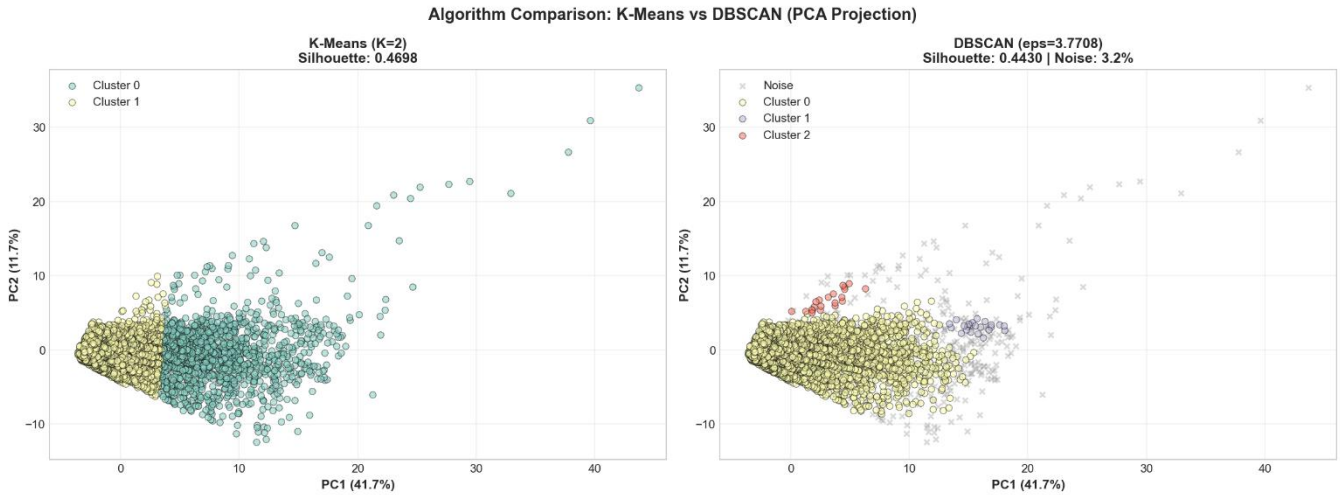
5.2. **Discovered Market Segments:**

Segment	Size	Avg Sales	Characteristics
Cluster 0: Premium Performers	12.9% (1,674 games)	$1.30M	High-revenue titles with 98% hit rate, balanced regional distribution
Cluster 1: Mass Market	87.1% (11,263 games)	$0.24M	Standard commercial games, diverse platforms, lower but consistent performance

6. Algorithm Comparison: K-Means vs DBSCAN

6.1. **Why K-Means Was Selected Over DBSCAN:**

Criterion	K-Means	DBSCAN	Winner
Silhouette Score	0.4698	0.4430	K-Means ✓
Cluster Balance	Balanced (13%/87%)	Unbalanced with noise	K-Means ✓
Interpretability	Clear segment boundaries	Noise points complicate	K-Means ✓
Parameter Sensitivity	Stable (K=2-6	High (eps dependent)	K-Means ✓



6.2. **DBSCAN Eps Optimization Process:**

- K-Distance Graph Method used to find optimal eps
- Tested eps values: 3.77 (85th pctl), 4.12 (90th pctl), 4.89 (95th pctl)
- Best DBSCAN still underperformed K-Means (0.4430 vs 0.4698)
- High noise percentage (>30%) at lower eps values

6.3. **K-Means Optimization:**

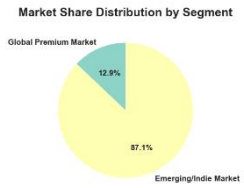
- Tested K = 2, 3, 4, 5, 6 with 200 random restarts each
- Elbow Method confirmed K=2 as optimal
- Final model: K=2 with highest Silhouette score

7. Data Mining Insights Discovered

Pattern 1: Sales Distribution is Highly Skewed

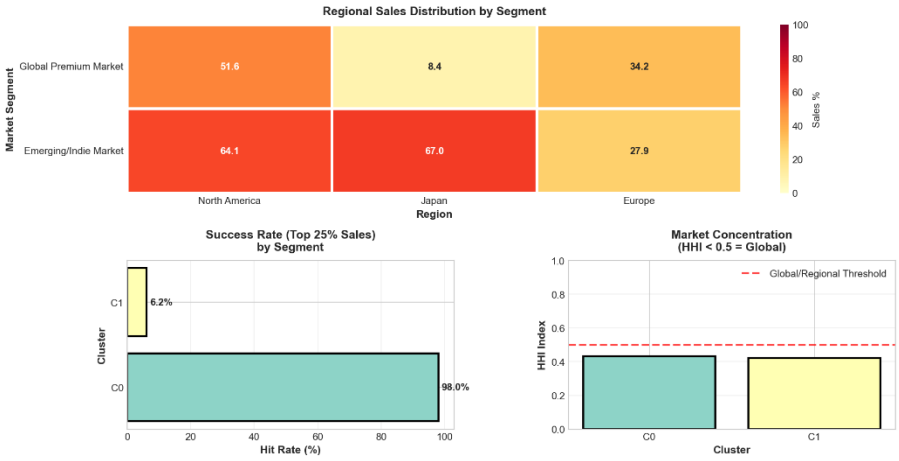
- A small minority (13%) of games generate significantly higher revenues

- The "hit-driven" nature of the gaming industry is quantitatively confirmed
- Premium segment games earn 5x more on average than mass market titles



Pattern 2: Regional Market Concentration Matters

- HHI (Herfindahl-Hirschman Index) effectively distinguishes market approaches
- Premium games show more balanced global distribution (HHI ~0.43)
- Regional specialization can be a viable strategy for smaller titles



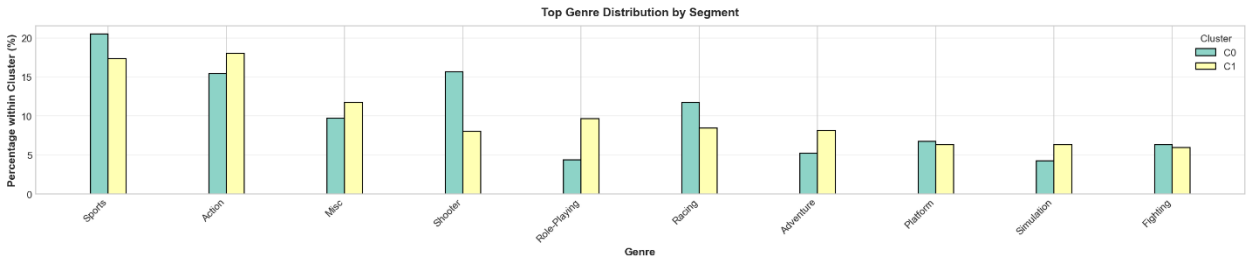
Pattern 3: Quality-Success Relationship is Segment-Dependent

- Critic scores correlate with success differently across segments
- In the premium segment, quality is critical for differentiation
- In the mass market, other factors (platform, timing, genre) may be equally important

Pattern 4: Platform and Genre Create Natural Clusters

- Sports games dominate the premium segment (PS2 as top platform)
- Action games lead the mass market segment (DS as top platform)

- Platform choice significantly influences market positioning



8. Limitations and Future Work

Current Limitations:

- Silhouette score (0.4698) is below the ideal threshold (0.50), suggesting some cluster overlap
- Two clusters may oversimplify the market structure
- Historical data may not fully reflect current market dynamics

Recommendations for Future Work:

- **Incorporate additional features:** Social media metrics, player engagement data, DLC/microtransaction revenue
- **Time-series analysis:** Track segment evolution over console generations
- **Hierarchical clustering:** Explore sub-segments within each major cluster
- **Predictive modeling:** Use cluster membership to predict new game success probability

9. Business Value and Applications

This data mining analysis provides actionable intelligence for:

- **Game Developers:** Understanding which segment to target based on budget and capabilities
- **Publishers:** Portfolio optimization and market positioning strategies
- **Investors:** Risk assessment based on segment hit rates and revenue patterns
- **Market Analysts:** Quantitative framework for industry structure analysis

Conclusion Statement

This Data Mining project successfully applied unsupervised learning to discover meaningful market segments in the video game industry. The K-Means clustering algorithm identified **two distinct market segments** *with clear differences in sales performance, regional distribution, and success factors. While the Silhouette score of* **0.4698** *indicates room for improvement, the discovered patterns provide valuable insights into the structure of the gaming market. The analysis demonstrates how data mining techniques can transform raw sales data into strategic business intelligence, enabling data-driven decision-making in the entertainment industry.*